

The below definitions are not ones you need to learn but will give you a sense of what the terms in italics mean as you will hear me use them a lot from now on. The idea of expectation is not actually on your syllabus but is a useful way of thinking about random variables and hence is included here.

- A *random variable* is a function mapping events in some sample space to some (usually numerical quantity). For example, we can define  $X$  to be the random variable denoting the sum of the two outcomes when two fair dice are rolled. The random variable is then effectively converting each event into a number, so if two fours are rolled, the random variable outputs an 8.
- The *probability distribution* of a random variable is the set of all possible (mutually exclusive) outcomes of the random variable with their associated probabilities.

### PROBLEMS:

1. The table below shows the probability distribution of the random variable  $X$ .

|            |               |               |     |               |                |
|------------|---------------|---------------|-----|---------------|----------------|
| $x$        | -2            | -1            | 0   | 1             | 2              |
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{5}$ | $k$ | $\frac{2}{5}$ | $\frac{1}{10}$ |

Find the value of the constant  $k$ .

2. A random variable  $X$  has a probability distribution given by

$$\mathbb{P}(X = r) = \begin{cases} kr^2 & \text{if } r = 1, 2, 3, 4; \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the value of  $k$ .

3. Let  $H$  model the highest common factor of the scores when two dice are rolled.
- Find the probability distribution of  $H$ .
  - Find  $P(H = 1 | H < 3)$ .

4. The probability distribution of a random variable  $X$  is given in the table below:

|            |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|
| $x$        | 0   | 1   | 2   | 3   | 4   |
| $P(X = x)$ | 0.3 | 0.2 | 0.1 | 0.1 | 0.3 |

The independent random variables  $X_1$  and  $X_2$  have the same distribution as  $X$ . The random variable  $Y = X_1 + X_2$ .

- Find  $P(Y = 8)$ .
  - Find  $P(Y = 1)$ .
  - Find  $P(Y = 6)$ .
5. The random variable  $X$  takes the values 1, 2, 3, or 4. The probability that  $X$  takes the value  $x$  is given by the formula

$$P(X = x) = \frac{1}{24}(x^2 - x + 1) \text{ for } x = 1, 2, 3, 4.$$

- Verify that the probabilities of the events  $X = 1$ ,  $X = 2$ ,  $X = 3$  and  $X = 4$  are all positive, and sum to 1.
  - Find the probability that  $X$  is even.
6. Let  $D$  model the difference between the numbers which show when two fair dice are rolled. Find the probability distribution of  $D$ .

7. The table below shows the probability distribution of the random variable  $Y$ .

|            |     |     |     |
|------------|-----|-----|-----|
| $y$        | 1   | 2   | 3   |
| $P(Y = y)$ | 0.2 | 0.3 | 0.5 |

Another random variable,  $Z$ , is independent of  $Y$ . The probability distribution for  $Z$  is shown in the table

|            |     |      |      |
|------------|-----|------|------|
| $z$        | 1   | 2    | 3    |
| $P(Z = z)$ | 0.1 | 0.25 | 0.65 |

One value of  $Y$  and one value of  $Z$  are chosen at random. Find the probability that

- $Y + Z = 3$ ,
- $Y \times Z$  is even.

8. The probability distribution of a random variable  $X$  is given in the table below:

|            |      |      |      |      |      |      |
|------------|------|------|------|------|------|------|
| $x$        | 0    | 1    | 2    | 3    | 4    | 5    |
| $P(X = x)$ | 0.13 | 0.23 | 0.09 | 0.35 | 0.12 | 0.08 |

Three independent random variables  $X_1, X_2, X_3$  each have the same distribution as  $X$ . The random variable  $Y = X_1 + X_2 + X_3$ .

- Find  $P(Y = 0)$ .
- Find  $P(Y = 1)$ .
- Find  $P(Y = 6)$ .

9. The random variable  $X$  can have the values 1, 2, 4 or 8. The probability that  $X$  has the value  $x$  is given by the formula  $P(X = x) = \frac{8}{15x}$  for  $x = 1, 2, 4, 8$ .
- Write down  $P(X = 3)$ .
  - Two independent values of  $X$  are obtained. Find the probability that their sum is at least 7.
10. The random variable  $X$  has the distribution  $P(X = x) = kx^2$ , for  $x = 1, 2, 3, 4, 5$ , where  $k$  is a constant. Two values of  $X$  are obtained independently. Determine the probability that their sum is 6.
11. A fair die is rolled until the first score of six is obtained.  $X$  models the number of rolls required. Find a formula for  $P(X = k)$ .